# MATH 144 Fall 2026

## Exam 1 Review

### Question 1 - Data & Study Design

A prominent investment firm based in the US wants to evaluate the portfolio management efficiency of its financial analysts. To conduct the study, 150 financial analysts were randomly selected from the firm's workforce. Among those sampled, 65 were classified as junior analysts, having less than five years of experience, while the remaining 85 were classified as senior analysts, with five or more years of experience. The analysts were responsible for managing investments across three key market sectors: technology, energy, and financial services.

The firm's primary focus was on how quickly analysts were able to generate a profitable return on a trade. On average, junior analysts required 3 days to close a profitable trade, whereas senior analysts were able to do so in about 2 days on average. Beyond trade execution speed, the firm also gathered additional information on each analyst, including their gender (male/female), their highesl level of education, and the average total value (in millions of dollars) of the portfolios under their management.

a. Classify this scenario as an *observational* or *experimental* study. Give a reason for your classification.

b. Identify the *sample* and the *population* of interest.

c. What are the *cases/observational units* in this study and how many are they?

d. Identify any one parameter and its associated statistic in this study, stating the specific values for each if given.

e. List ***all*** the variables in the study and identify each variable as numerical or categorical. For each numerical variable state whether it is discrete or continuous. For each categorical variable state whether it is nominal or ordinal.

f. Identify a variable (or variable set if more than one) that can be reasonably visualized using:

- A simple bar plot
- A simple histogram
- Simple box plot
- Side-by-side box plots
- Side-by-side bar plots
- Scatter plot

## Question 2 - Data Summarization

Below are weights (in pounds) of 6 randomly sampled patients who visited St. Joseph's Clinic on a certain day.

$$182, 186, 177, 185, 183, 181$$

a. Find the **5-number summary** for the weights then sketch a box plot to visualize their distribution.

b. Use the $1.5 \times IQR$ rule to determine if there are any suspected outliers among the six weights.

b. Calculate the *standard deviation* for the weights and interpret its meaning. Show/explain your work.

c. Suppose a nurse notices an error in the data entry. The weight entered as 186 lbs should have been 168 lbs. Without doing any calculations, state whether each of the following statistics will increase, decrease, or stay the same if the 186 is changed to 168. Explain your reasoning.

- mean

- median

- standard deviation

- interquartile range

- Range

## Question 3 - Hypothesis Testing

A business analyst wants to perform statistical inference to answer the research question,

> Do arrival delay rates (proportions) differ between Delta and American Airlines?.

The `Airlines` data set from the `Stat2Data` package contains information on over 10,000 flights by Delta and American Airlines arriving at LaGuardia and O'Hare airports. These data are summarized as shown in the table below:

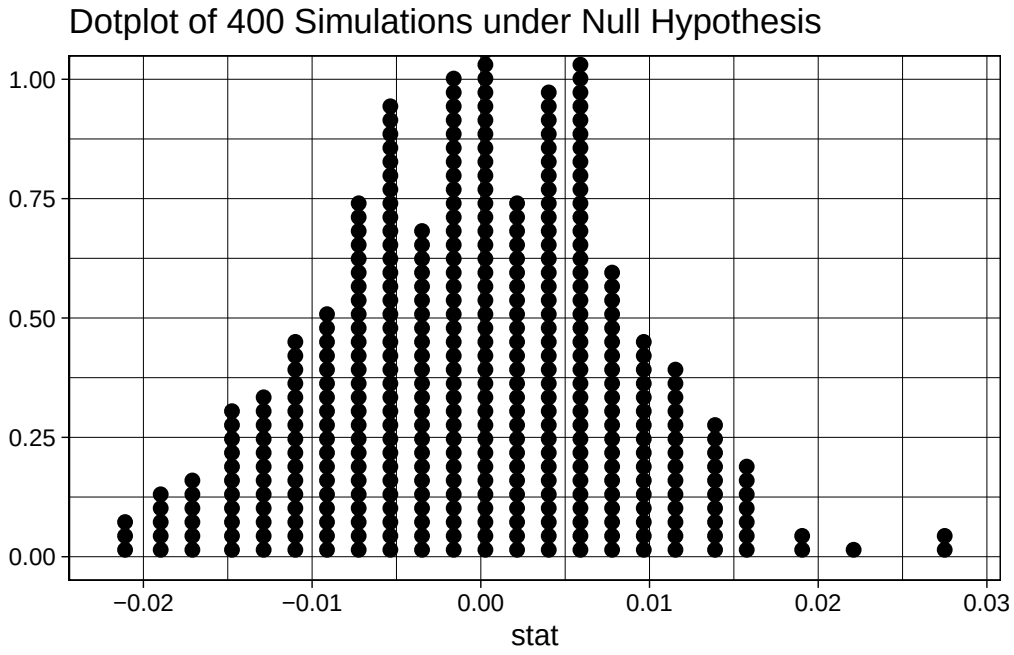
| Airline | Delayed Arrivals | On-Time Arrivals | Total |
|---|---|---|---|
| Delta | 659 | 2344 | 3003 |
| American | 1450 | 5880 | 7330 |
| Total | 2109 | 8224 | 10333 |

a. Identify the predictor and response variable in the above scenario.

b. State the null and alternative hypotheses in words and in symbols.

c. Calculate the observed statistic (i.e., the difference in delay proportions in the sample). Does the observed statistic suggest a difference in delay rates between the two airlines?

d. The histogram below shows the distribution of 400 simulated statistics under the null hypothesis.

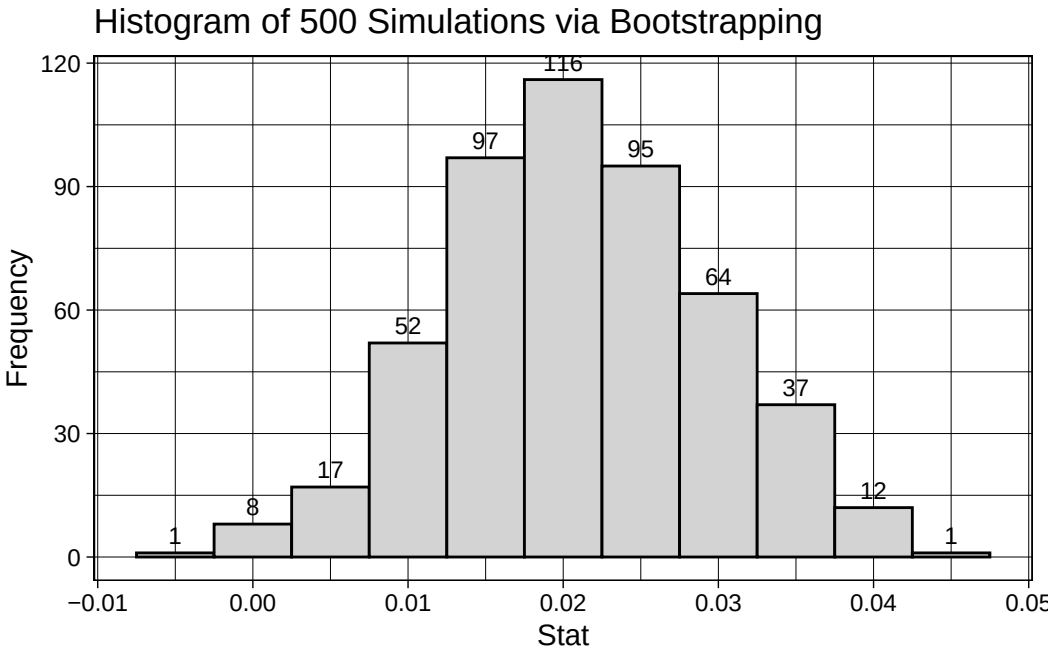



- Explain why this distribution is centered around 0.

- Use the dot plot to estimate the *p-value* (show/explain work) then write your conclusion for the hypothesis test in context.

## Question 4 - Confidence Interval

The histogram below shows the distribution of 400 statistics (differences in proportion of delays) generated by bootstrapping using the airlines data in problem #3 above (check out class activities if you need a refresher on bootstrapping). The number on top of each bin shows the frequency of statistics that fall within that bin and each bin is 0.005 units wide.




a. Explain why this distribution is centered around 0.2.

b. Use the histogram to estimate the 95% confidence interval. Show/explain your work.

c. Interpret the confidence interval in context.

d. State whether the interval you found supports the conclusion reached under the hypothesis testing part (question 3d). Explain.